Warm Up

1.) Find $\frac{dy}{dx}$ if $y = \ln\left(\frac{x^2\sin(x)}{2x+1}\right)$. (*Hint*: *Don't* use the Chain Rule. Instead, use properties of logarithms *before* differentiating.)

2.) Let $y = \frac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}$. Does the previous example give you any clever idea for finding $\frac{dy}{dx}$? Explain.

Logarithmic Differentiation

To differentiate $y = h(x)$:

1. Take the natural logarithm of both sides to obtain $\ln(y) = \ln(h(x))$.
2. Use properties of logarithms to expand $\ln(h(x))$ as much as possible.
3. Differentiate both sides of the equation. On the left we will have $\frac{1}{y}\frac{dy}{dx}$.
4. Multiply both sides of the equation by y to solve for $\frac{dy}{dx}$.
5. Replace y by $h(x)$.

3.) Find the derivative of $y = \dfrac{x^{3/4}\sqrt{x^2+1}}{(3x+2)^5}$.

4.) Find the derivative of $y = \dfrac{x\sqrt{2x+1}}{e^x \sin^3(x)}$.

5.) Can you use the Power Rule or the Exponential Rule to find the derivative of $y = x^x$? Explain why or why not and find $\frac{dy}{dx}$.

6.) Find the derivative of $y = x^{\sqrt{x}}$.

7.) Find the derivative of $y = (2x^4 + 1)^{\tan(x)}$.

We can now use logarithmic differentiation to prove the general version of the Power Rule.

Theorem (General Power Rule). If α is any real number and $f(x) = x^\alpha$, then

$$f'(x) = \alpha x^{\alpha-1} \quad \text{wherever it exists.}$$

8.) Before proving this theorem, explain why logarithmic differentiation fails in finding the derivative of $y = x^3$ for nonpositive values of x.

We can take care of the negative values of x by means of the following formula:

9.) Show that $\frac{d}{dx}\ln(|x|) = \frac{1}{x}$.

If some values of $y = h(x)$ are negative, then we simply take the absolute value of both sides of this equation before using logarithmic differentiation, using the formula above for $\frac{d}{dx}\ln(|x|)$.

10.) Prove the general version of the Power Rule for all nonzero values of x using the modification of logarithmic differentiation discussed above. [1] (*Hint:* You may use without proof that $|x^{\alpha}| = |x|^{\alpha}$ for any real number α.)

11.) Show that if $f(x) > 0$ and $g(x) > 0$, then the derivative of $y = f(x)^{g(x)}$ is given by an expression that looks something like the sum of the Exponential Rule and the Power Rule.

[1] For $x = 0$, one can show that $\frac{d}{dx}x^{\alpha}|_{x=0} = 0$ for $\alpha \geq 1$ (and DNE otherwise) by using the definition of the derivative.